



What's your take on fuel cell based vehicles and hybrid vehicles? Would these be better than pure EVs?

Hybrid has been a proven technology as compared to EV and at the same time more fuel-efficient than IC-engine, especially in the eastern part of the world. However, the cost of ownership is much higher for a hybrid vehicle than an IC-engine vehicle. This could be the mid-way path for some OEMs. It will give us time to address the scarcity of charging stations and other infrastructure, which are a bit of a bottleneck right now. But Mahindra and Tata are confident that EV is the way forward.

Do you believe that we will see hybrid two-wheelers too?

There will be hybrid 2-wheelers, but electric 2-wheelers will take the lead. There is already lots of investment by private players in electric 2-wheelers. The government is also supporting this industry with lots of tax benefits to the manufacturers as well as the consumers.

Is there any role for the government in terms of standardisation?

Government needs to play a major role in standardisation and other activities. To begin with, we need to have a standard EV policy across all states. There is a big disparity within the country, in terms of charge per unit for the charging stations. Clear guidelines and targets are needed for faster roll-out of charging stations across the country, especially on the highways. Government should direct the government-owned petrol companies to install more charging stations on their premises. Banks need to be encouraged to provide soft loans. All this will encourage private players also to participate. And if battery-as-a-service sort of thing catches up, more and more entrepreneurs will

jump into it because there is a lot of money.

Is there a need to standardise the EV chargers too?

Yes, there is an urgent need for it. An accessible and affordable charging infrastructure will be the greatest enabler for EV infrastructure. BIS needs to help reduce multiple charging standards that should comprehensively cover the type of connectors used, power, and voltage specifications.

What are your views on battery swapping?

For 2- and 3-wheelers, battery swapping is the best. It offers upfront reduction in the cost of vehicle. Their battery size being small, they are easy to swap. Battery-as-a-service can be a new business model. The space constraint for setting up charging stations in urban areas can be addressed by battery swapping. But 4-wheelers, commercial vehicles, and buses will continue to require charging stations within the cities and on highways. So, there's a market for both swapping as well as charging stations in India.

What percentage of EVs can we expect in the future?

If you look at the government's roadmap, and if I remember the numbers correctly, by 2030 almost 80% of 2- and 3-wheelers are expected to be electric. EV 2-wheeler is the major focus and then next comes commercial vehicles up to 70% of the available market in India.

With a large chunk of our population living in small towns and villages, are we going to see similar charging technologies in both the cities and in the villages, or will there be a difference?

My take is that the EVs will evolve gradually. When we say 80% of 2-wheelers becoming electric, I'm

looking at the rural population also, which will be a major contributor. The rural population may adopt home charging and the urban may prefer battery swapping or a faster charging technology.

Do you foresee any technology coming up that would make charging at home faster?

Fast charging of cars and larger vehicles at home will not happen. In our housing complexes, most transformers are rated at 20kW. In a medium-size housing society having, say, 400 apartments, charging about 15 vehicles in one go on a fast charger is still fine as the transformer may be able to provide sufficient power on an average day. But, if we go beyond this number, the present residential power infrastructure will not suffice.

What are the essential features a BMS must have to ensure the safety of the vehicle and avoid thermal runaway?

As per me, the ideal BMS needs to handle: (a) cell balancing to maintain equal voltage levels, (b) thermal control to prevent thermal runaway due to high external temperature, (c) notification of voltage, current, and temperature levels of the battery, and (d) notification of the charge remaining. All these should be conveyed through a BMS dashboard and diagnosed remotely.

It is being said that the EV batteries may have to undergo Conformity of Production (CoP) testing. What are your views?

This is still in its initial stages. It covers the battery components, such as cells. With such standards in place, it will be possible to penalise an OEM for non-confirming to the standards.

More importantly, it would give more confidence to the consumers to buy the electric vehicles. **EFY**